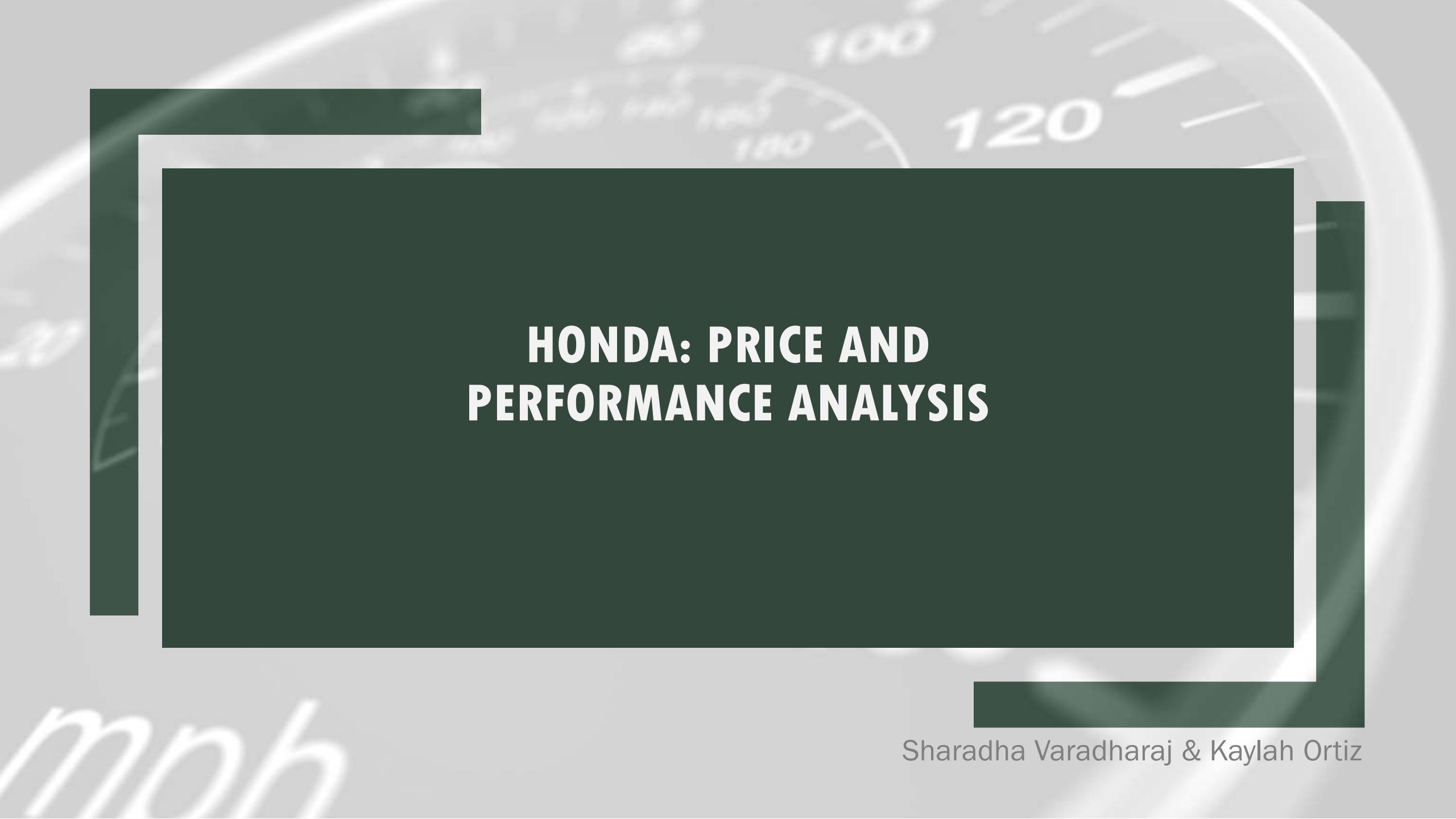




HONDA: PRICE AND PERFORMANCE ANALYSIS

Sharadha Varadharaj & Kaylah Ortiz

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2. ANOVA
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Our data

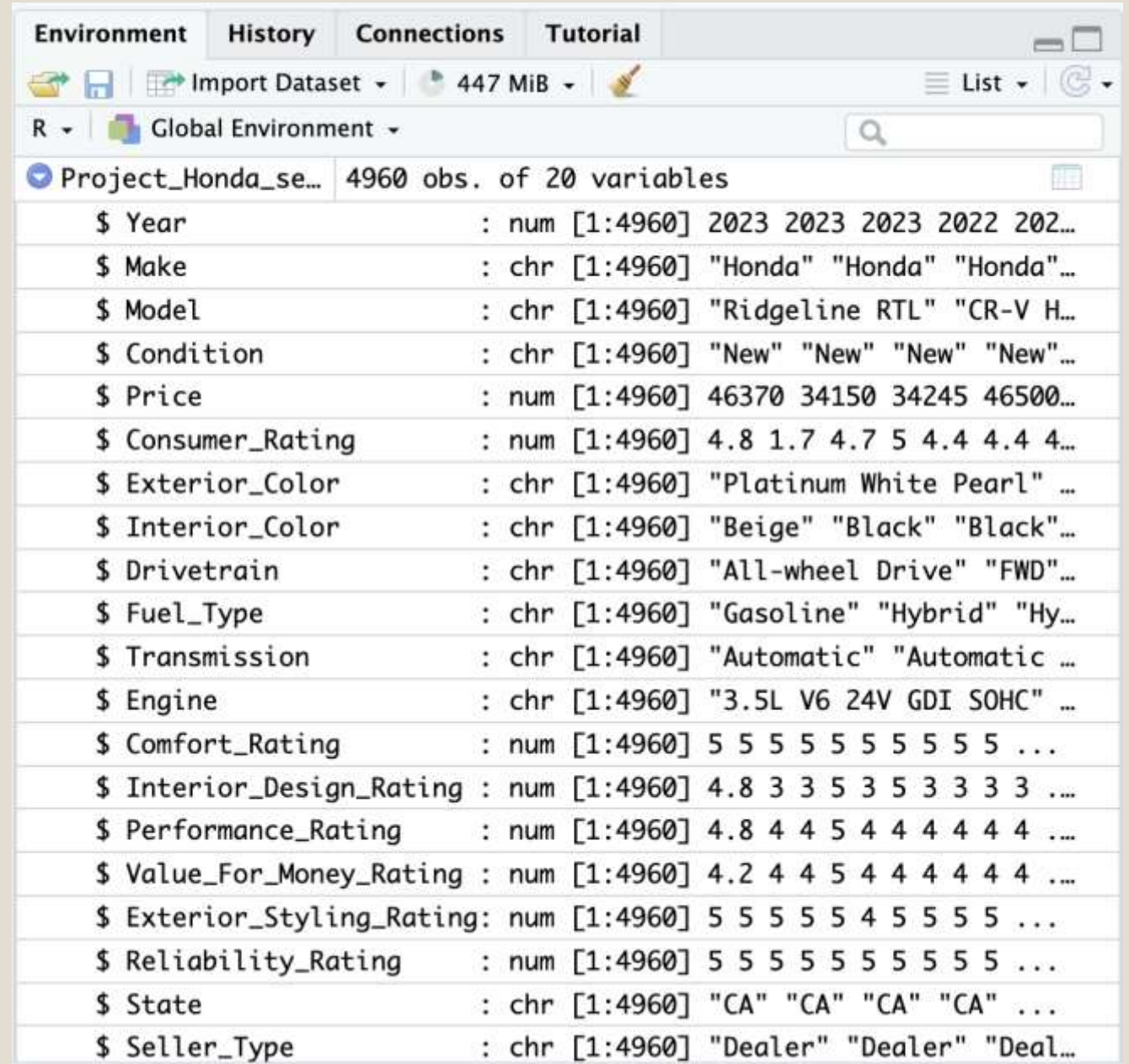
4960 observations

Response Variable:
Price

Predictor Variable:
19 variables



NOTE: Human-Survey Data
=
Human Bias



Environment	History	Connections	Tutorial
R Global Environment			
Project_Honda_se... 4960 obs. of 20 variables			
\$ Year	: num [1:4960]	2023 2023 2023 2022 202...	
\$ Make	: chr [1:4960]	"Honda" "Honda" "Honda"...	
\$ Model	: chr [1:4960]	"Ridgeline RTL" "CR-V H..."	
\$ Condition	: chr [1:4960]	"New" "New" "New" "New"...	
\$ Price	: num [1:4960]	46370 34150 34245 46500...	
\$ Consumer_Rating	: num [1:4960]	4.8 1.7 4.7 5 4.4 4.4 4...	
\$ Exterior_Color	: chr [1:4960]	"Platinum White Pearl" ...	
\$ Interior_Color	: chr [1:4960]	"Beige" "Black" "Black"...	
\$ Drivetrain	: chr [1:4960]	"All-wheel Drive" "FWD"...	
\$ Fuel_Type	: chr [1:4960]	"Gasoline" "Hybrid" "Hy..."	
\$ Transmission	: chr [1:4960]	"Automatic" "Automatic ...	
\$ Engine	: chr [1:4960]	"3.5L V6 24V GDI SOHC" ...	
\$ Comfort_Rating	: num [1:4960]	5 5 5 5 5 5 5 5 5 5 ...	
\$ Interior_Design_Rating	: num [1:4960]	4.8 3 3 5 3 5 3 3 3 3 ...	
\$ Performance_Rating	: num [1:4960]	4.8 4 4 5 4 4 4 4 4 4 ...	
\$ Value_For_Money_Rating	: num [1:4960]	4.2 4 4 5 4 4 4 4 4 4 ...	
\$ Exterior_Styling_Rating	: num [1:4960]	5 5 5 5 5 4 5 5 5 5 ...	
\$ Reliability_Rating	: num [1:4960]	5 5 5 5 5 5 5 5 5 5 ...	
\$ State	: chr [1:4960]	"CA" "CA" "CA" "CA" ...	
\$ Seller_Type	: chr [1:4960]	"Dealer" "Dealer" "Deal..."	

Research Question

Which factors contribute most to the reasoning of the pricing of Honda cars?

Data cleaning and analysis

We removed 6-Speed M/T from the Fuel Type predictor

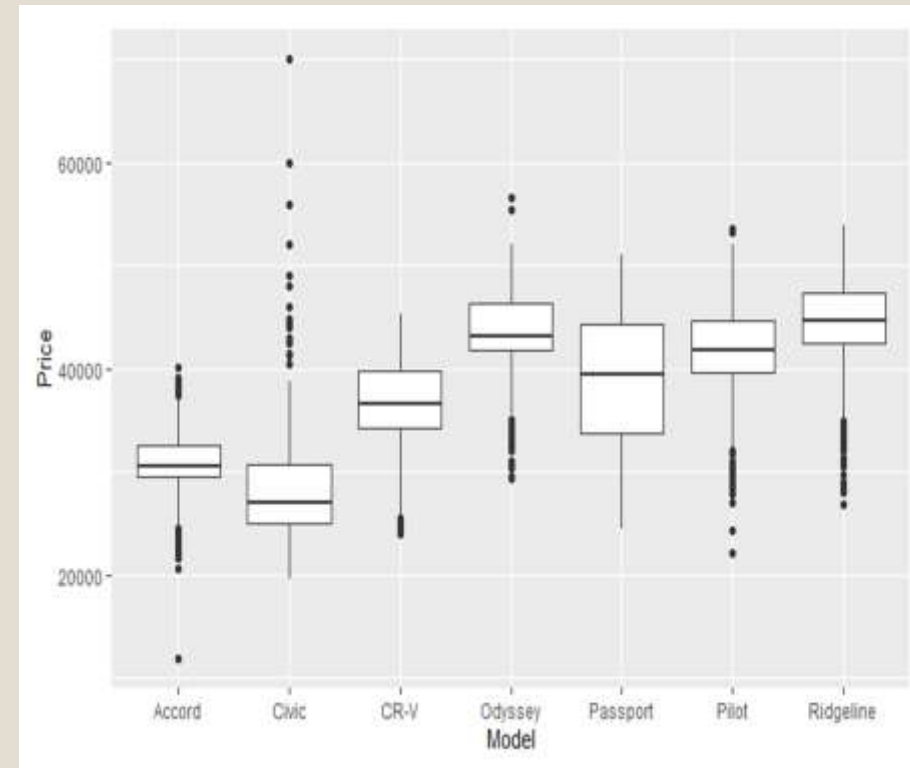
```
Fuel_Type|
table(AP_ProjectData_Honda$Fuel_Type)

6-Speed M/T  Gasoline  Hybrid
1           2516     414
```

Other cleaning:

- We condensed the Model and States of the original dataset
- No missing data
- Condensed Models into 7 types

Future Problems?



Combining and Removing variables with Multivariate Model

- Removed few variables for not being significant in the model
- We realized that we should combine 2 of our models within the Model category
 - *Created “Other” category within Model*
 - *Accord and Civic due to their close estimates within the model*
 - *Made all our factors significant*

Multivariate Model

Final Multivariate Model

- Resulted in all p-values being significant

```
model1.lm = lm(Price ~ relevel(as.factor(Model), ref = 3)  
               + as.factor(Year)  
               + Value_For_Money_Rating  
               + Performance_Rating,  
               data = Honda3)  
summary(model1.lm)
```

ANOVA testing

- Helpful for concluding that an interaction is not significant if only one category of our categorical variables were insignificant in interaction testing

Interaction testing

- Interaction testing resulted in 3 interactions that came out significant
- We put all 3 in the model
 - *only one came out significant* <- VFM:PR

```
model1.lm = lm(Price ~ relevel(as.factor(Model), ref = 3)
               + as.factor(Year)
               + Performance_Rating * Value_For_Money_Rating
               + Value_For_Money_Rating
               + Performance_Rating,
               data = Honda3)
summary(model1.lm)
```

OUTPUT



```
Call:
lm(formula = Price ~ relevel(as.factor(Model), ref = 3) + as.factor(Year) +
    Performance_Rating * Value_For_Money_Rating + Value_For_Money_Rating +
    Performance_Rating, data = Honda3)

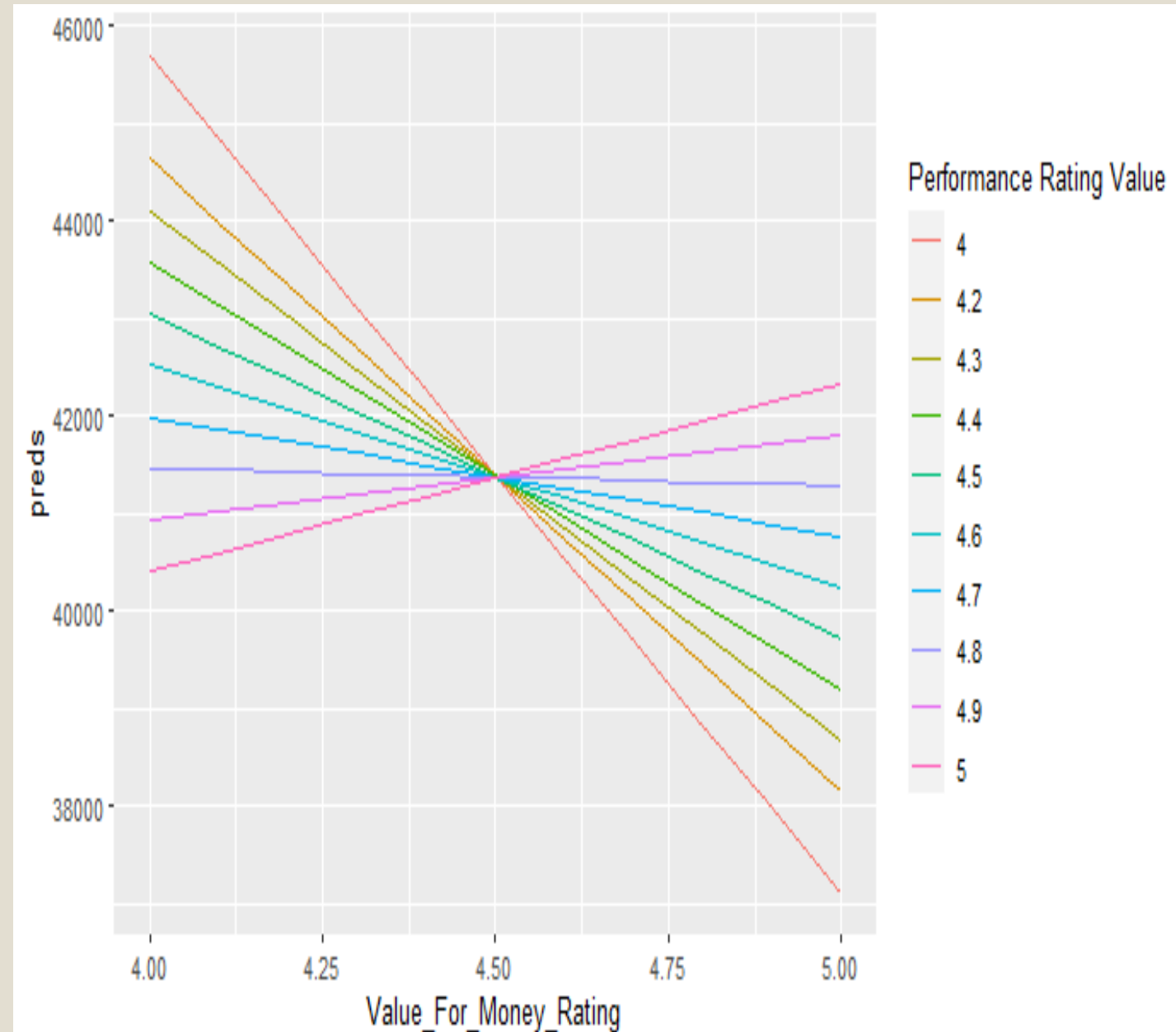
Residuals:
    Min       1Q   Median       3Q      Max
-15228  -2504    -51     1812   41403

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    221674.1    23690.8   9.357 < 2e-16 ***
relevel(as.factor(Model), ref = 3)1    11526.4     337.5  34.153 < 2e-16 ***
relevel(as.factor(Model), ref = 3)2     4521.1     292.0  15.482 < 2e-16 ***
relevel(as.factor(Model), ref = 3)4    12731.6     327.9  38.830 < 2e-16 ***
relevel(as.factor(Model), ref = 3)5     9892.4     272.8  36.257 < 2e-16 ***
relevel(as.factor(Model), ref = 3)6    10230.3     205.2  49.844 < 2e-16 ***
as.factor(Year)2021         2645.0      299.5   8.831 < 2e-16 ***
as.factor(Year)2022         6182.8      267.9  23.075 < 2e-16 ***
as.factor(Year)2023         9723.9      396.8  24.507 < 2e-16 ***
Performance_Rating    -43903.9      4686.1  -9.369 < 2e-16 ***
Value_For_Money_Rating  -46340.9      5774.9  -8.025 1.46e-15 ***
Performance_Rating:Value_For_Money_Rating  10258.9      1147.0   8.944 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3656 on 2918 degrees of freedom
Multiple R-squared:  0.716,    Adjusted R-squared:  0.7149
F-statistic: 668.7 on 11 and 2918 DF,  p-value: < 2.2e-16
```

Interpretation

- We have evidence ($p < .05$) to suggest that there is a significant interaction effect between Performance_Rating and Value_For_Money_Rating. The result suggests treating the lines as not parallel.



Our Final Model

- We ended with 4 factors and 1 interaction remaining within our final model:

```
model1.lm = lm(Price ~ relevel(as.factor(Model), ref = 3)
               + as.factor(Year)
               + Performance_Rating * Value_For_Money_Rating
               + Value_For_Money_Rating
               + Performance_Rating,
               data = Honda3)
summary(model1.lm)
```

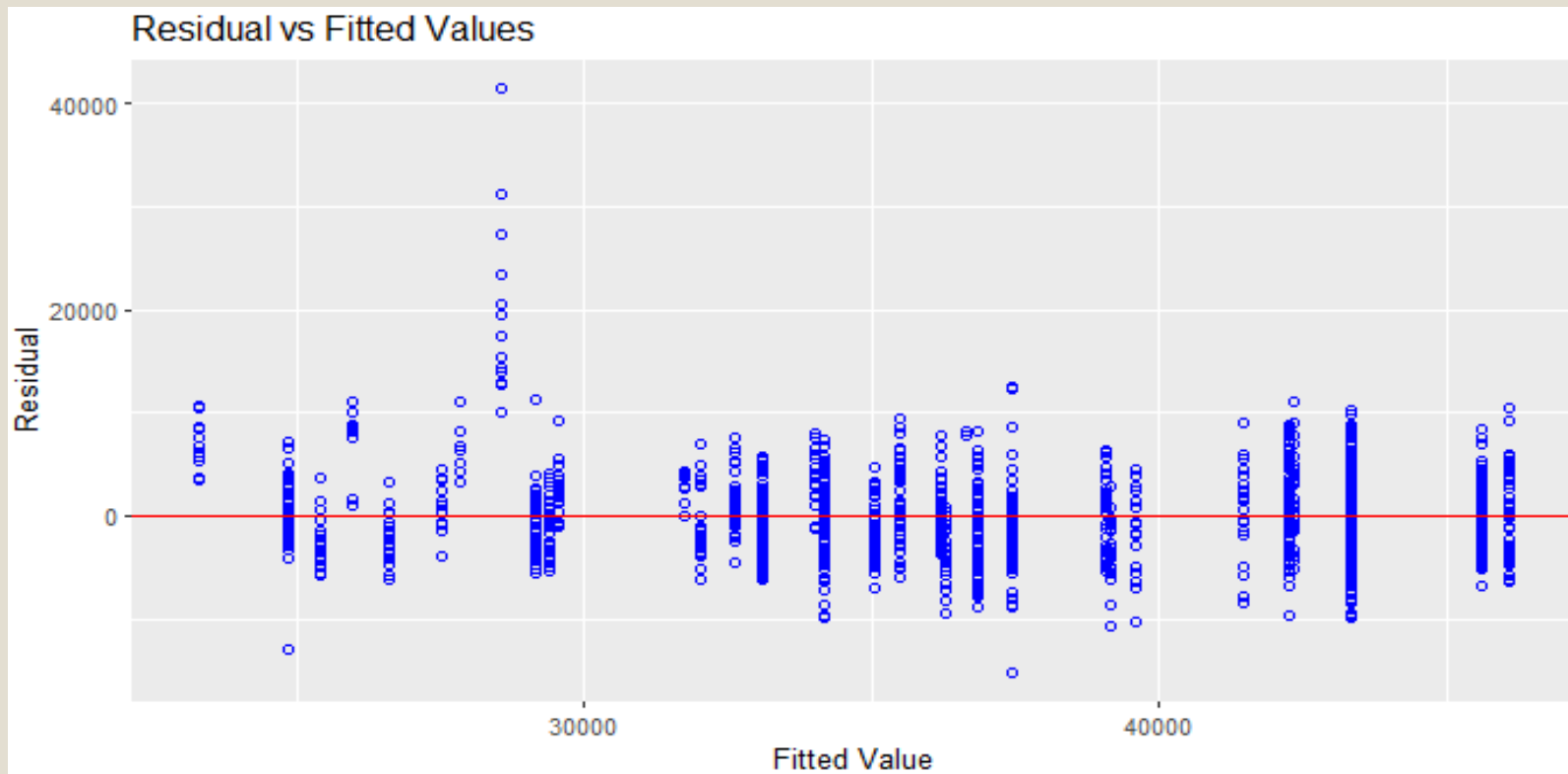
LINE Testing

- Linearity
- Independence
- Normality
- Equal Variance

LINE Testing

Linearity Assumption

- Slight outliers but not much of concern since no true pattern present
- Since all categorical variables, no requirement of transformation



LINE Testing

Independence Assumption

- Durbin Watson test < 2
- Runs test = insignificant

```
> # Durbin-Watson: No AutoCorrelation
> library(car)
> durbinWatsonTest(model1.lm)
lag Autocorrelation D-W Statistic p-value
1      0.02153363      1.956905    0.242
Alternative hypothesis: rho != 0
> # Runs Test
> library(tseries)
> runs.test(factor(sign(rstandard(model1.lm))))
```

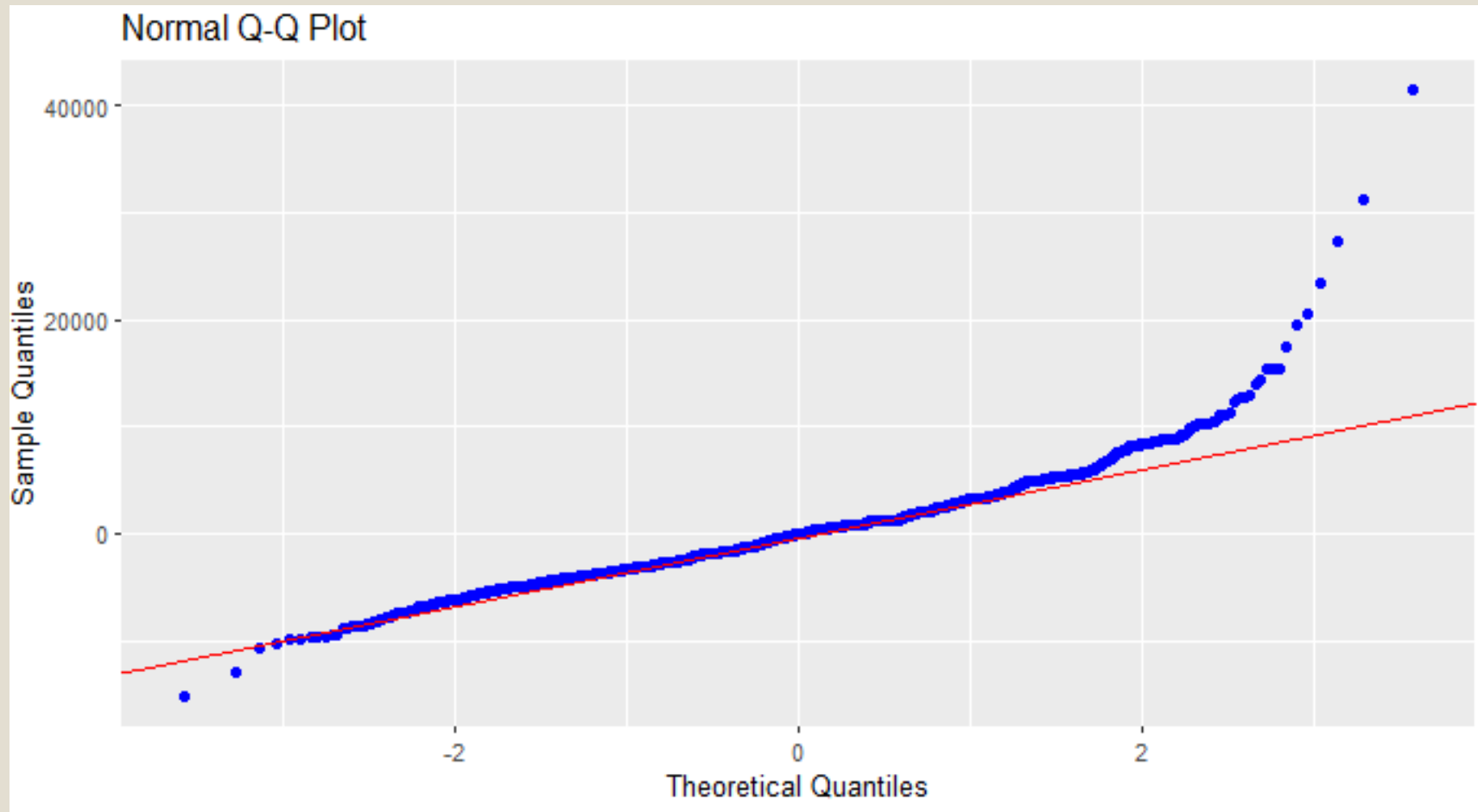
Runs Test

```
data: factor(sign(rstandard(model1.lm)))
Standard Normal = 0.95015, p-value = 0.342
alternative hypothesis: two.sided
```

LINE Testing

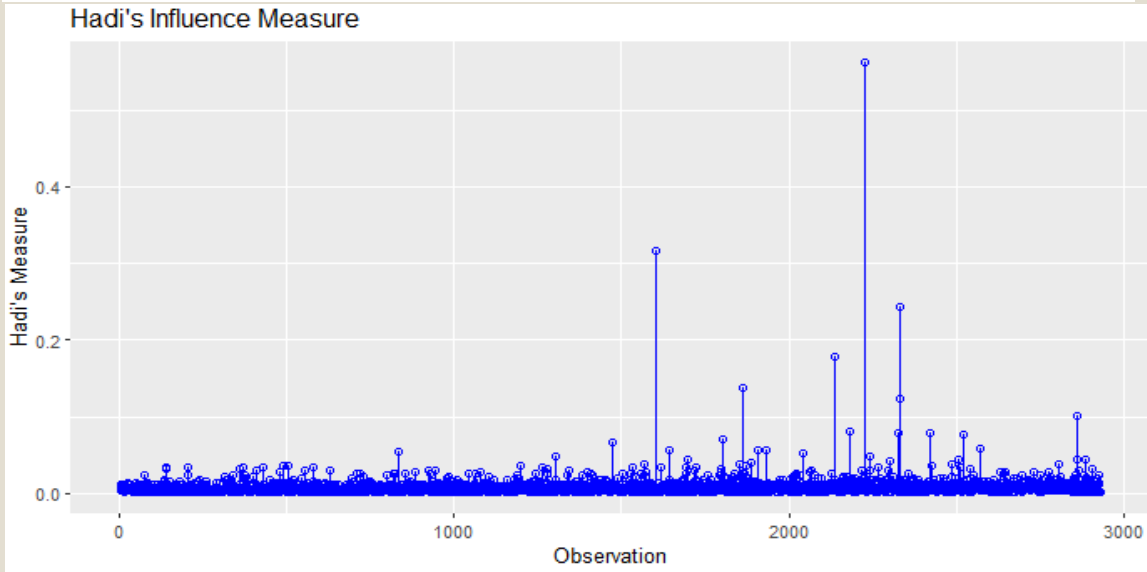
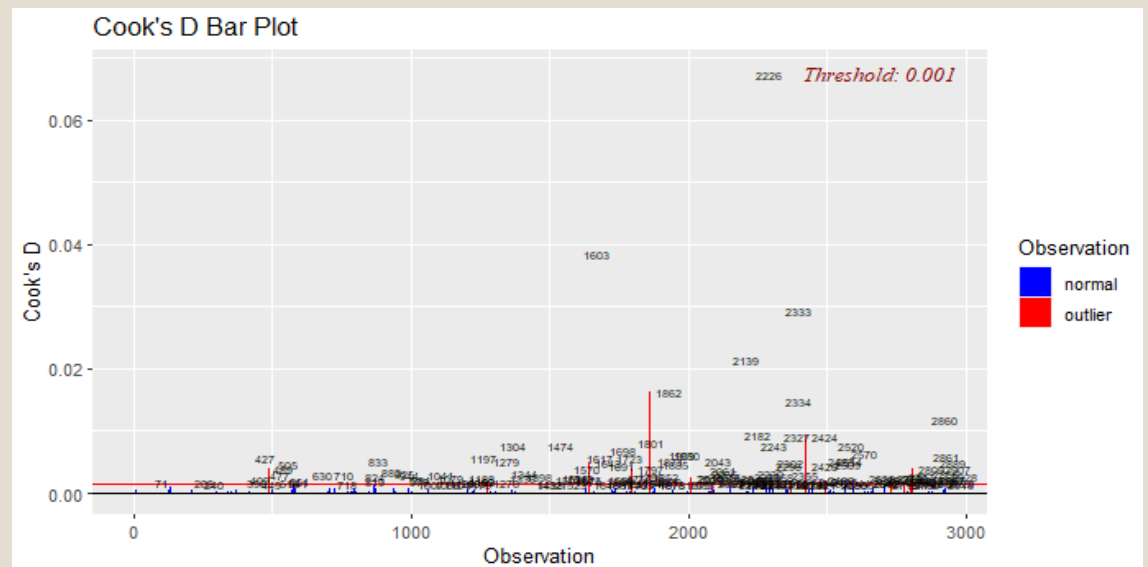
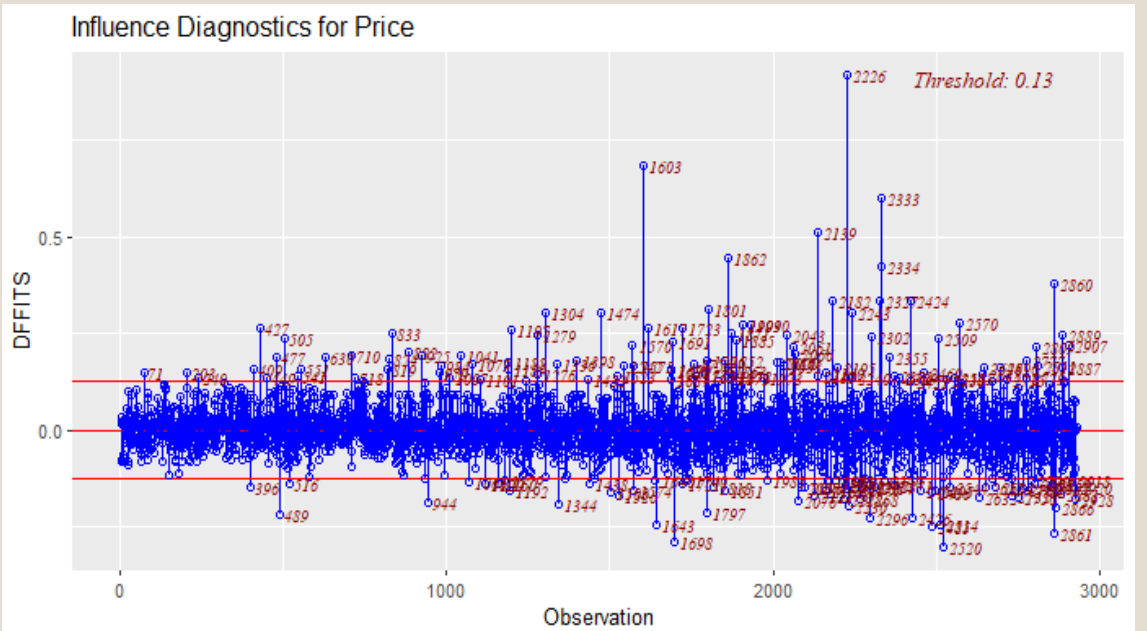
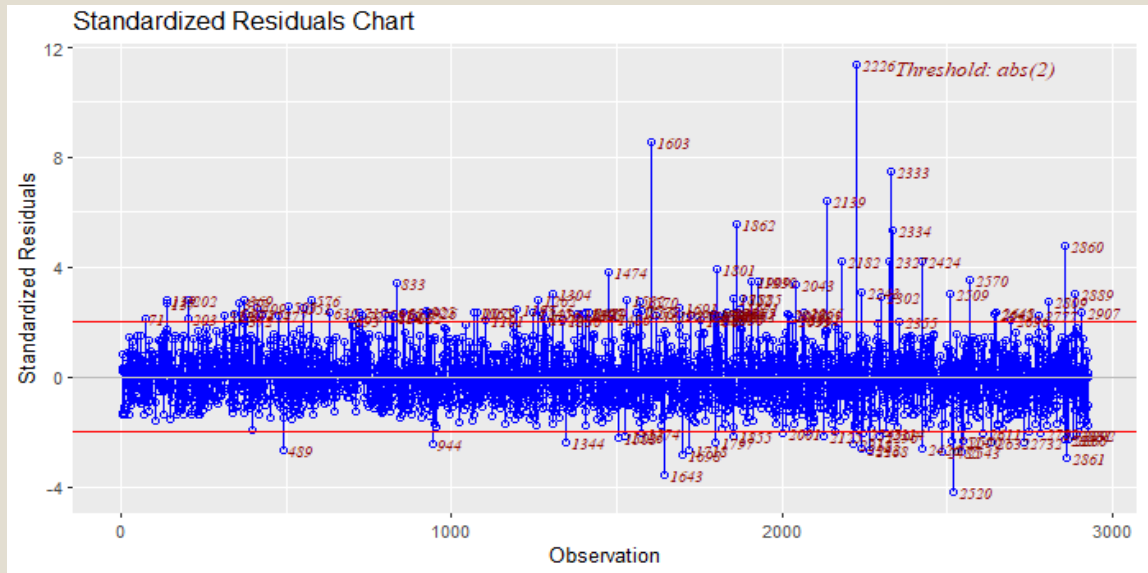
Normality Assumption

- Majority of the points do fit on the linear line
- No requirement of transformations



Influential Points

Point: 2226



Refitting the Model

- Taking out the influential point fit the model again

```
reduce_data <- Honda3[-2226, ]

reduce_data.lm = lm(Price ~ relevel(as.factor(Model), ref = 3)
  + Year
  + Performance_Rating * Value_For_Money_Rating
  + Value_For_Money_Rating
  + Performance_Rating,
  data = Honda3)

summary(reduce_data.lm)
summary(model1.lm)
```

Refitting the Model

Before removing influential point

```
Call:
lm(formula = Price ~ relevel(as.factor(Model), ref = 3) + Year +
    Performance_Rating * Value_For_Money_Rating + Value_For_Money_Rating +
    Performance_Rating, data = Honda3)

Residuals:
    Min       1Q   Median       3Q      Max
-15228  -2504    -51    1812   41403

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
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relevel(as.factor(Model), ref = 3)1    11526.4     337.5  34.153 < 2e-16 ***
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relevel(as.factor(Model), ref = 3)6    10230.3     205.2  49.844 < 2e-16 ***
Year2021         2645.0      299.5   8.831 < 2e-16 ***
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F-statistic: 668.7 on 11 and 2918 DF,  p-value: < 2.2e-16
```

After removing influential point

```
Call:
lm(formula = Price ~ relevel(as.factor(Model), ref = 3) + Year +
    Performance_Rating * Value_For_Money_Rating + Value_For_Money_Rating +
    Performance_Rating, data = reduce_data)

Residuals:
    Min       1Q   Median       3Q      Max
-15117.5  -2517.1     21.7    1785.5   31528.7

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    212621.0    23178.2   9.173 < 2e-16 ***
relevel(as.factor(Model), ref = 3)1    11585.3     330.0  35.103 < 2e-16 ***
relevel(as.factor(Model), ref = 3)2     4568.7     285.6  15.999 < 2e-16 ***
relevel(as.factor(Model), ref = 3)4    12782.8     320.6  39.868 < 2e-16 ***
relevel(as.factor(Model), ref = 3)5     9979.2     266.9  37.391 < 2e-16 ***
relevel(as.factor(Model), ref = 3)6    10315.4     200.8  51.365 < 2e-16 ***
Year2021         2500.0      293.1   8.528 < 2e-16 ***
Year2022         6225.9      262.0  23.761 < 2e-16 ***
Year2023        9774.9      388.0  25.193 < 2e-16 ***
Performance_Rating -42104.2    4584.7  -9.184 < 2e-16 ***
Value_For_Money_Rating -44106.0    5650.0  -7.806 8.14e-15 ***
Performance_Rating:Value_For_Money_Rating 9809.4    1122.2   8.741 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3575 on 2917 degrees of freedom
Multiple R-squared:  0.7265,    Adjusted R-squared:  0.7255
F-statistic: 704.6 on 11 and 2917 DF,  p-value: < 2.2e-16
```

Insights

Value for Money Rating and Performance Rating:

- Due to the negative estimates for VFM and PR, with Price being the predictor variable (Inverse relationship), we can assume that customers might take affordability into consideration when looking at prices of cars.

Year:

- Due to the positive estimates for Year we can conclude that car prices are probably increasing over time.
 - Maybe because of inflation, market, or other economic factors.

Reliability of model

- Low standard error
- P-values significant
- R squared value is 71%
 - *We can explain 71% of the variance in Price collectively explained by our predictors*

Strengths and limitations

- Troubles:

- Started off with a lot of variables and 10+ categories within each variable
- Interaction Testing knowledge

- Strengths:

- No transformations (Residual Plot)

Further research

- Regarding Performance Ratings, we could produce further analysis to understand what consumers find important to them regarding the car in terms of “Performance”.
 - *Examples: Smoothness of drive, MPG, safety features, equipped for families, tire type*
- With this information, Honda could address needs of both niche or generalized customer pools to produce new models that cater towards their consumers better.

Further research

■ Economic Factors:

- *Economic conditions, inflation, fuel prices, electric cars in the market*
 - Possible reason for year increasing over time

■ Competitor Factors:

- *Identify factors that contribute to differences in prices with groups of similar car types of different brands*

Questions?